Homework 06

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$$

$$a_k = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T}\int_T x(t)\, e^{-jk(2\pi/T)t}\, dt$$

(a) Find a_k of the Fourier Series for the periodic waveforms of (1) and (2).

(b) Calculate the value a_k $(-10 \le k \le 10)$ for the periodic waveforms of (1) and (2).

(c) Draw bar graphs for (1) and (2) for the periodic waveforms of (1) and (2).

(d) The coefficients are same as the coefficients obtained in the Homework 05?